

Euclid's Elements — Book XI

Proposition 16

Formal Reconstruction — Technical Documentation

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If two parallel planes be cut by a plane, then their common sections are parallel straight lines.

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1 Introduction

Proposition XI.16 is the basic section theorem for parallel planes. If two planes π and ρ are parallel and a third plane σ meets both of them, then the two section lines cut out in σ are parallel.

Classically the argument is very short. The sections are straight lines by Book XI, and if those lines met in a point P , then P would belong to both π and ρ , contradicting the assumption that the two planes are parallel.

The formal reconstruction preserves exactly this idea, but in a more explicit incidence form. The theorem does not start with already named section lines. Instead, it begins with two witness points

$$A \in \pi \cap \sigma, \quad B \in \rho \cap \sigma,$$

constructs the corresponding intersection lines, and then proves that those lines are parallel.

This makes XI.16 an important architectural proposition in Book XI. It is genuinely spatial, because three ambient planes are involved, but it is also conceptually clean: the entire production proof lives in the incidence layer. No order, no congruence, and no Euclidean parallel axiom are used in the geometric core.

2 Classical Configuration

Let π and ρ be parallel planes. Let a third plane σ cut them. Write

$$l = \pi \cap \sigma, \quad m = \rho \cap \sigma.$$

Euclid's claim is that the two section lines l and m are parallel.

In the formal theorem the two intersections are represented through explicit points:

$$A \in \pi \cap \sigma, \quad B \in \rho \cap \sigma.$$

These witnesses guarantee that σ actually meets each of the two reference planes.

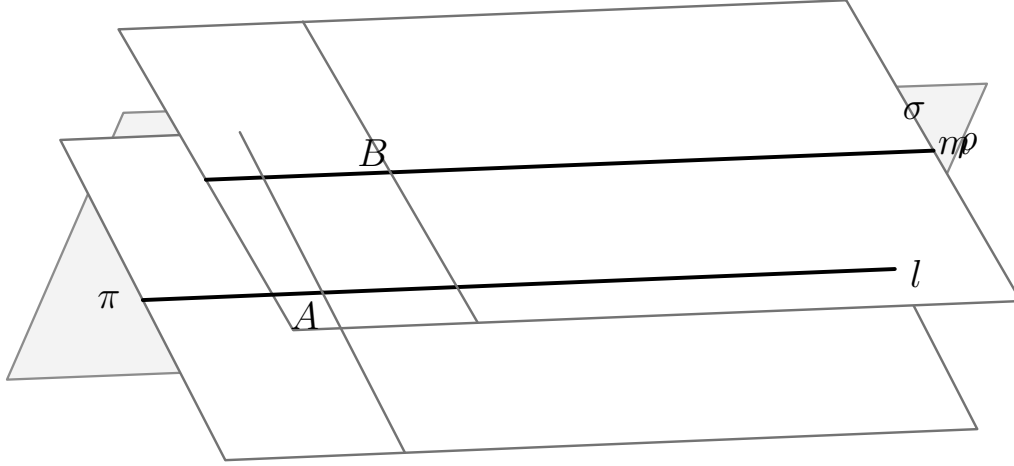


Figure 1: A spatial rendering of the configuration of Proposition XI.16. The displayed reference plane is σ . The two parallel section lines $l = \pi \cap \sigma$ and $m = \rho \cap \sigma$ lie in σ and are drawn in black. The ambient planes π and ρ are shown as light auxiliary planes cutting the reference plane along those lines.

Diagrammatic audit. A single figure is sufficient here. The proposition has only one stable geometric state: two parallel ambient planes cut by a third plane. The proof contains no separate degenerate branch and no later planar reduction requiring an additional diagram.

3 Statement

Theorem 1 (Euclid XI.16). *If two parallel planes are cut by a plane, then their common sections are parallel straight lines.*

The current production declaration is:

```

theorem euclid_proposition_11_16
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (pi rho sigma : S.Plane)
  (A B : Geo.Point)
  (hPlanesParallel :
    HilbertSpacePlanesParallelIncidence Geo pi rho)
  (hApi : S.OnPlane A pi)
  (hAsigma : S.OnPlane A sigma)
  (hBrho : S.OnPlane B rho)
  (hBsigma : S.OnPlane B sigma) :
  exists l m : Geo.Line,

```

```

H.OnLine A l /\
H.OnLine B m /\
(forall X : Geo.Point,
  (S.OnPlane X pi /\ S.OnPlane X sigma) <->
  H.OnLine X l) /\
(forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X sigma) <->
  H.OnLine X m) /\
HilbertSpaceLinesParallel Geo l m

```

Two features should be emphasized.

First, the public theorem is not stated with pre-existing section lines. Instead it *constructs* the two lines l and m from the data of the plane intersections.

Second, the theorem records the intersections pointwise. For every point X , lying on the section line is equivalent to lying in both relevant planes. This is the precise formal replacement for the informal notation

$$l = \pi \cap \sigma, \quad m = \rho \cap \sigma.$$

4 Classical Proof

Let the common sections of π and ρ with σ be the straight lines l and m .

Assume, for contradiction, that l and m meet at a point P . Since $P \in l$ and l is the section of π with σ , the point P lies in π . Since $P \in m$ and m is the section of ρ with σ , the point P also lies in ρ .

Thus the two planes π and ρ have a common point. But parallel planes do not meet. Therefore the lines l and m do not meet. Because both lie in the same plane σ , they are parallel.

The classical dependency pattern is therefore

$$\begin{array}{c}
\pi \parallel \rho, \quad l = \pi \cap \sigma, \quad m = \rho \cap \sigma \\
\downarrow \\
\text{assume } l \cap m \neq \emptyset \\
\downarrow \\
P \in \pi \cap \rho \\
\downarrow \\
\perp \\
\downarrow \\
l \parallel m.
\end{array}$$

The proof is neutral: no Euclidean parallel postulate and no metric argument enter.

5 What is genuinely three-dimensional here?

The conclusion concerns two lines in the single plane σ , but the proposition is still genuinely spatial.

The genuinely three-dimensional data are:

- the two ambient planes π and ρ ;
- the cutting plane σ ;
- the two ambient intersections $\pi \cap \sigma$ and $\rho \cap \sigma$;
- the contradiction step that turns a hypothetical meeting point of the section lines into a common point of the two ambient planes.

What is striking in XI.16 is what *does not* happen. There is no reduction to $\text{PlaneGeo}(\sigma)$, no use of planar order, and no appeal to planar congruence or angle theory. The proposition stays entirely in the ambient incidence layer.

6 Formal Statement and Parallelism Predicate

The new neutral infrastructure for parallel planes is

```
def HilbertSpacePlanesParallelIncidence
  [S : HilbertSpacePrimitive Geo]
  (pi rho : S.Plane) : Prop :=
  Not
    (exists X : Geo.Point,
      S.OnPlane X pi /\
      S.OnPlane X rho)
```

Thus the meaning of

$$\pi \parallel \rho$$

in the production proof is simply: the two planes have no common point.

The target parallelism for lines is the existing spatial predicate

```
HilbertSpaceLinesParallel Geo l m
```

which packages two facts:

- the two lines lie in a common plane;
- the two lines are disjoint.

In XI.16 the common plane is exactly the cutting plane σ .

7 Proof Architecture of the Reconstruction

The formal proof has four stable stages.

7.1 Stage 1: show that the intersections are nontrivial

To apply the general intersection theorem for two planes, the proof must first show

$$\pi \neq \sigma, \quad \rho \neq \sigma.$$

These facts are not hypotheses of the theorem. They are derived from the parallel-plane assumption by contradiction.

If $\pi = \sigma$, then the point B , already known to lie in $\rho \cap \sigma$, would lie in both π and ρ , contradicting parallelism. The same argument with A gives $\rho \neq \sigma$.

7.2 Stage 2: construct the two section lines

With those inequalities available, the proof applies the ambient incidence theorem

```
hilbert_plane_intersection_line
```

twice:

- to π and σ , using the point A ;
- to ρ and σ , using the point B .

This yields lines l and m , together with:

- $A \in l$,
- $B \in m$,
- $l \subseteq \pi$ and $l \subseteq \sigma$,
- $m \subseteq \rho$ and $m \subseteq \sigma$,
- the pointwise equivalences

$$X \in l \iff X \in \pi \cap \sigma, \quad X \in m \iff X \in \rho \cap \sigma.$$

This is the key formal replacement for the informal geometric phrase “the common section of two planes.”

7.3 Stage 3: prove that the two section lines are disjoint

Assume a point P lies on both l and m . Because

$$l \subseteq \pi, \quad m \subseteq \rho,$$

the point P lies in both π and ρ . But this contradicts

```
hPlanesParallel :
  HilbertSpacePlanesParallelIncidence Geo pi rho
```

So the lines l and m are disjoint.

7.4 Stage 4: assemble spatial line parallelism

The final step is immediate. Both l and m lie in σ , and they are disjoint. Therefore they are parallel in the sense of

```
HilbertSpaceLinesParallel Geo 1 m
```

This closes the theorem.

8 Wyler-Style Flat Reconstruction

The independent Wyler reconstruction of XI.16 keeps exactly the same geometric content but changes the preferred formal state. Instead of storing the two plane intersections only as pointwise equivalences, it records them as exact equalities of carriers.

Write

$$C(\pi) = \text{carrier}(\pi), \quad C(l) = \text{carrier}(l).$$

For the first section line the Wyler proof obtains

$$C(\pi) \cap C(\sigma) = C(l),$$

and for the second one

$$C(\rho) \cap C(\sigma) = C(m).$$

In Lean these identities are expressed through

```
HilbertPlaneCarrier3D
HilbertLineCarrier3D
hilbertPlaneCarrier3D_inter_eq_lineCarrier
```

The last theorem is a flat-facing wrapper around the neutral three-dimensional plane-intersection theorem. Thus the Wyler version does not introduce a different incidence theorem. It changes the level at which the result is retained: the intersection is preserved as a reusable *meet identity*.

The proof first derives

$$\pi \neq \sigma, \quad \rho \neq \sigma$$

from the parallel-plane hypothesis, exactly as in the canonical proof. It then constructs l and m and records

$$C(l) = C(\pi) \cap C(\sigma), \quad C(m) = C(\rho) \cap C(\sigma).$$

These identities immediately recover all four containments

$$l \subseteq \pi, \quad l \subseteq \sigma, \quad m \subseteq \rho, \quad m \subseteq \sigma$$

whenever they are needed.

The decisive disjointness argument becomes especially transparent in this language. If a point P belonged to both section lines, then

$$P \in C(l) \cap C(m).$$

Using the two meet identities gives

$$P \in C(\pi) \cap C(\sigma) \cap C(\rho) \cap C(\sigma),$$

hence in particular

$$P \in C(\pi) \cap C(\rho),$$

contradicting the parallelism of the original planes. Therefore

$$C(l) \cap C(m) = \emptyset.$$

The final line-parallelism statement still uses the ordinary spatial predicate

```
HilbertSpaceLinesParallel Geo l m
```

because both section lines lie in the common cutting plane σ and are disjoint there.

The complete Wyler-style structure is therefore

```
pi || rho
|
sigma meets pi and rho
|
+--> C(pi) inter C(sigma) = C(l)
|
+--> C(rho) inter C(sigma) = C(m)
|
assume P in C(l) inter C(m)
|
P in C(pi) inter C(rho)
|
contradiction
|
C(l) inter C(m) = empty
|
l,m subset sigma
|
l || m
```

The production theorem is

```
euclid_proposition_11_16_wyler
```

and it coexists with the canonical theorem because the Wyler route now has its own public name.

For XI.16 the difference between the two reconstructions is therefore not a difference in mathematical strength. The canonical proof stores a pointwise description of the two intersections; the Wyler proof stores exact carrier equalities. The latter is more compositional for subsequent spatial arguments, because a later theorem may rewrite directly with the meet identity instead of reconstructing membership information point by point. This is precisely why the Wyler form becomes useful in XI.17.

9 Lean Formalization

The production proof is notably small because the geometry is concentrated in the two applications of the plane-intersection constructor. The theorem begins by deriving

```
have hPiSigma : Ne pi sigma := ...
have hRhoSigma : Ne rho sigma := ...
```

Then it constructs the first section line:

```
cases
  hilbert_plane_intersection_line
    (Geo := Geo)
    pi sigma
    hPiSigma
    A hApi hAsigma with
| intro l hLData =>
```

and similarly for the second one:

```
cases
  hilbert_plane_intersection_line
    (Geo := Geo)
    rho sigma
    hRhoSigma
    B hBrho hBsigma with
| intro m hMData =>
```

The contradiction argument is then expressed as:

```
have hDisjoint :
  HilbertLinesDisjoint Geo l m := by
  intro hMeet
  cases hMeet with
| intro P hPData =>
  have hP1 : H.OnLine P l := hPData.1
  have hPm : H.OnLine P m := hPData.2
  apply hPlanesParallel
  refine Exists.intro P ?_
  exact And.intro (hlpi P hP1) (hmrho P hPm)
```

Finally, line parallelism is assembled in the common plane σ .

10 Hidden Formal Data

Euclid's statement suppresses several obligations that Lean must make explicit.

- The cutting plane does not arrive with named section lines; the proof must construct them.

- The theorem needs witness points A and B showing that σ actually meets the two parallel planes.
- The inequalities

$$\pi \neq \sigma, \quad \rho \neq \sigma$$

are required before the plane-intersection theorem can be used.

- “Common section” must be encoded formally by a universal characterization of points on the line.
- Parallelism of lines is not primitive equality of directions; it is a structured statement combining coplanarity with disjointness.

These are not artificial complications. They are the formal content hidden by the classical diagram.

11 Relationship with Earlier Book XI Propositions

Classically, XI.16 belongs to the cluster of early Book XI results about plane sections and parallelism. In textbook language one would naturally refer back to the fact that the common section of two planes is a straight line.

The current formal reconstruction, however, does not import any earlier Book XI production theorem. Instead it uses the general ambient incidence theorem

```
hilbert_plane_intersection_line
```

from the 3D Hilbert infrastructure.

So the mathematical content agrees with Euclid, but the formal dependency DAG is shorter at the Book XI level and deeper at the infrastructure level.

12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No proposition from Book I is used here, and there is no dependence on the Book Zero segment infrastructure.

The proof lives entirely in the 3D incidence layer:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
```

The key infrastructure items are:

```
HilbertSpacePlanesParallelIncidence
```

```

hilbert_plane_intersection_line
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel

```

This makes XI.16 one of the cleanest examples in Book XI of a theorem whose formal content is purely spatial incidence.

13 Dependency Summary

Classical.

$$\begin{array}{c}
\pi \parallel \rho, \quad \sigma \text{ cuts } \pi, \rho \\
\downarrow \\
l = \pi \cap \sigma, \quad m = \rho \cap \sigma \\
\downarrow \\
\text{if } l \cap m \neq \emptyset, \text{ then } \pi \cap \rho \neq \emptyset \\
\downarrow \\
\perp \\
\downarrow \\
l \parallel m.
\end{array}$$

Formal.

$$\begin{array}{c}
\text{Hilbert incidence in 3-space} \\
\downarrow \\
\text{HilbertSpacePlanesParallelIncidence} \\
\downarrow \\
hPiSigma, \quad hRhoSigma \\
\downarrow \\
\text{hilbert_plane_intersection_line twice} \\
\downarrow \\
l, \quad m, \quad \text{pointwise section characterizations} \\
\downarrow \\
\text{HilbertLinesDisjoint} \\
\downarrow \\
\text{HilbertSpaceLinesParallel} \\
\downarrow \\
\text{euclid_proposition_11_16.}
\end{array}$$

The conceptual difference is clear. Euclid speaks directly about the common sections. Lean first constructs them from the ambient incidence interface and then proves their disjointness.

14 Conclusion

Proposition XI.16 is mathematically elementary, but formally very informative. It shows that an early Book XI result can be reconstructed in a completely synthetic and purely incidence-theoretic way.

This theorem is also structurally important for later propositions. In particular, both the canonical and the Wyler reconstruction of XI.17 use XI.16 as their spatial section engine, while

the more delicate ratio argument of XI.17 is handled in a separate proportion layer. XI.16 therefore marks a clean boundary: it solves the ambient intersection geometry once and for all, without importing any metric or affine machinery. The Wyler version makes that boundary explicit by storing the two sections as exact carrier meets.